

Cholesterol crystals cause mechanical damage to biological membranes: a proposed mechanism of plaque rupture and erosion leading to arterial thromb...

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BACKGROUND: Plaque rupture and/or erosion is the leading cause of cardiovascular events; however, the process is not well understood. Although certain morphologic characteristics have been associated with ruptured plaques, these observations are of static histological images and not of the dynamics of plaque rupture. To elucidate the process of plaque rupture, we investigated the transformation of cholesterol from liquid to solid crystal to determine whether growing crystals are capable of injuring the plaque cap. HYPOTHESIS: We hypothesized that during cholesterol crystallization the spatial configuration rapidly changes, causing forceful expansion of sharp-edged crystals that can damage the plaque cap. METHODS: Two experiments were performed in vitro: first, cholesterol powder was melted in graduated cylinders and allowed to crystallize at room temperature. Volume changes from liquid to solid state were measured and timed. Second, thin biological membranes (20-40 microm) were put in the path of growing crystals to determine damage during crystallization. RESULTS: As cholesterol crystallized, the peak volume increased rapidly by up to 45% over 3 min and sharp-tipped crystals cut through and tore membranes. The amount of cholesterol and peak level of crystal growth correlated directly ($r = 0.98$; $p < 0.01$), as did the amount of cholesterol and rate of crystal growth ($r = 0.99$; $p < 0.01$). CONCLUSIONS: These observations suggest that crystallization of supersaturated cholesterol in atherosclerotic plaques can induce cap rupture and/or erosion. This novel insight may help in the development of therapeutic strategies that can alter cholesterol crystallization and prevent acute cardiovascular events.